ABIR ASHAB NILOY

Website: [https://abir-ashab.github.io] | [Google Scholar] | [GitHub] | [LinkedIn]
Email: [abirashabniloy@usf.edu]

EDUCATION

University of South Florida *Fall 2026 - present*
Ph.D. in Computer Science and Engineering
Advisor: Dr. Amit Seal Ami

Institute of Information Technology, University of Dhaka *2022-2026*
Bachelor of Science in Software Engineering

PROFESSIONAL EXPERIENCE

Academia

- **Graduate Teaching Assistant**, University of South Florida, Tampa, Florida *August 2026-present*

Industry

- **Associate Software Engineer** at Cefalo, Dhaka, Bangladesh *March 2025-June 2026*
- **Software Engineer Trainee** at Xpeedlab, Dhaka, Bangladesh *January 2024-March 2024*

PUBLICATIONS

- *Multi-Source Cybersecurity Logs: An ATT&CK-Labeled Dataset and SLM Evaluation*, **AA Niloy**, A. Ryan, I. H. Rafi, M. Erfan, and M. R. Rahman; arXiv preprint arXiv:2606.18190 *2026*

DATASETS AND PROJECTS

Multi-Source ATT&CK-Labeled Log Dataset for Attack Detection

- Built a multi-source (system, network, and browser) log dataset of 870 sessions (70 attack, 800 benign) and ∼2.3 million events, labeled with MITRE ATT&CK technique IDs spanning 12 tactics and 53 techniques.
- Emulated multi-stage attacks and collected telemetry using the ELK Stack to capture cross-source attack behavior.
- Fine-tuned three small language models (Qwen2.5-1.5B, Llama-3.2-3B, Phi-4-Mini) with LoRA, raising chunk-level technique classification accuracy from ∼8% to 90–97%.
- Code and data: github.com/Abir-Ashab/Mitre-Dataset

DocuMind: Smart Document Processing with Retrieval-Augmented Generation [Demo]

- Built a RAG-based system that transcribes meetings into notes, supports chat over documents, and generates quizzes from their content.
- Python, Whisper, ChromaDB, Gemini, React, Express

BanglaBridge: LLM-Powered Banglish-to-Bangla Translation [GitHub]

- Developed a platform that translates Banglish (Bengali written in Latin script) into Bangla, with an AI chatbot, content generation, and language-learning features built on LLMs.
- Ollama, Gemini, React, Node.js, MongoDB

TEACHING

Graduate Teaching Assistant at University of South Florida

- COT 4400 - Analysis of Algorithms *Fall 2026*
  - Grade homework assignments and exams, with written feedback on algorithm correctness and complexity analysis.
  - Hold office hours to help students work through problems in algorithm design, recurrences, and asymptotic analysis.
  - Meet with students individually to clarify course concepts and support their progress in the course.

RECOGNITIONS

Hackathons

- Second Runner-Up, AI Hackathon, Creative IT and *The Daily Star* *2025*
- Third Runner-Up and Best UI Award, Bitfest Hackathon, KUET *2025*
- Special Mention, Cefalo AI Hackathon *2025*
- Finalist (top 20 of 200+ teams), BUET Hackathon *2024*
- Finalist (top 10 of 150+ teams), DU ITVerse Hackathon, University of Dhaka *2023*

Competitive Programming

- Specialist on Codeforces (max rating 1441, 400+ problems solved)
- 3-star on CodeChef (max rating 1707)

REFERENCES

- **Dr. Amit Seal Ami**, Assistant Professor, University of South Florida (*Ph.D. Advisor*)
  amitsealami@usf.edu
- **Dr. Md Rayhanur Rahman**, Assistant Professor, The University of Alabama (*Research Advisor*)
  mdrayhanur.rahman@ua.edu
- **Dr. Kazi Muheymin-Us-Sakib**, Professor, IIT, University of Dhaka (*Undergraduate Instructor*)
  muheymin@gmail.com